

where P_0 is a 3×3 matrix. The recursive estimate of the three-element vector x is then obtained from Equation (3.47) as follows:

For $k = 1, 2, \dots$,

$$\begin{aligned} K_k &= P_{k-1} H_k^T (H_k P_{k-1} H_k^T + R_k)^{-1} \\ \hat{x}_k &= \hat{x}_{k-1} + K_k (y_k - H_k \hat{x}_{k-1}) \\ P_k &= (I - K_k H_k) P_{k-1} (I - K_k H_k)^T + K_k R_k K_k^T \end{aligned} \quad (3.68)$$

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3.4 WIENER FILTERING

In this section, we will give a brief review of Wiener filtering. The rest of this book does not assume any knowledge on the reader's part of Wiener filtering. However, Wiener filtering is important from a historical perspective, and it still has a lot of applications in signal processing and communication theory. But since it is not used much for state estimation anymore, the reader can safely skip this section if desired.

Wiener filtering addresses the problem of designing a linear, time-invariant filter to extract a signal from noise, approaching the problem from the frequency domain perspective. Norbert Wiener invented his filter as part of the World War II effort for the United States. He published his work on the problem in 1942, but it was not available to the public until 1949 [Wie64]. His book was known as the “yellow peril” because of its mathematical difficulty and its yellow cover [Deu65, page 176]. Andrey Kolmogorov actually solved a more general problem earlier (1941), and Mark Krein also worked on the same problem (1945). Kolmogorov's and Krein's work was independent of Wiener's work, and Wiener acknowledges that Kolmogorov's work predated his own work [Wie56]. However, Kolmogorov's and Krein's work did not become well known in the Western world until later, since it was published in Russian [Kol41]. A nontechnical account of Wiener's work is given in his autobiography [Wie56].

To set up the presentation of the Wiener filter, we first need to ask the following question: How does the power spectrum of a stochastic process $x(t)$ change when it goes through an LTI system with impulse response $g(t)$? The output $y(t)$ of the system is given by the convolution of the impulse response with the input:

$$y(t) = g(t) * x(t) \quad (3.69)$$

Since the system is time-invariant, a time shift in the input results in an equal time shift in the output:

$$y(t + \alpha) = g(t) * x(t + \alpha) \quad (3.70)$$

Multiplying the above two equations and writing out the convolutions as integrals gives

$$y(t)y(t + \alpha) = \int g(\tau)x(t - \tau) d\tau \int g(\gamma)x(t + \alpha - \gamma) d\gamma \quad (3.71)$$

Taking the expected value of both sides of the above equation gives the autocorrelation of $y(t)$ as a function of the autocorrelation of $x(t)$:

$$E[y(t)y(t + \alpha)] = \int \int g(\tau)g(\gamma)E[x(t - \tau)x(t + \alpha - \gamma)] d\tau d\gamma \quad (3.72)$$

which we will write in shorthand notation as

$$R_y(\alpha) = \int \int g(\tau)g(\gamma)R_x(\alpha + \tau - \gamma) d\tau d\gamma \quad (3.73)$$

Now we take the Fourier transform of the above equation to obtain

$$\int R_y(\alpha)e^{-j\omega\alpha} d\alpha = \int \int \int g(\tau)g(\gamma)R_x(\alpha + \tau - \gamma)e^{-j\omega\alpha} d\tau d\gamma d\alpha \quad (3.74)$$

Now we define a new variable of integration $\beta = \alpha + \tau - \gamma$ and replace α in the above equation to obtain

$$\begin{aligned} S_y(\omega) &= \int \int \int g(\tau)g(\gamma)R_x(\beta)e^{-j\omega\beta}e^{-j\omega\gamma}e^{j\omega\tau} d\tau d\gamma d\beta \\ &= G(-\omega)G(\omega)S_x(\omega) \end{aligned} \quad (3.75)$$

In other words, the power spectrum of the output $y(t)$ is a function of the Fourier transform of the impulse response of the system, $G(\omega)$, and the power spectrum of the input $x(t)$.

Now we can state our problem as follows: Design a stable LTI filter to extract a signal from noise. The quantities of interest in this problem are given as

$$\begin{aligned} x(t) &= \text{noise free signal} \\ v(t) &= \text{additive noise} \\ g(t) &= \text{filter impulse response (to be designed)} \\ \hat{x}(t) &= \text{output of filter [estimate of } x(t)] \\ e(t) &= \text{estimation error} \\ &= x(t) - \hat{x}(t) \end{aligned} \quad (3.76)$$

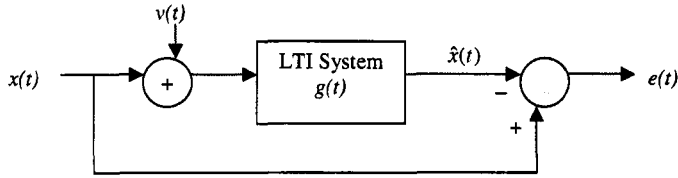


Figure 3.2 Wiener filter representation.

These quantities are represented in Figure 3.2, from which we see that

$$\begin{aligned} \hat{x}(t) &= g(t) * [x(t) + v(t)] \\ \hat{X}(\omega) &= G(\omega)[X(\omega) + V(\omega)] \\ E(\omega) &= X(\omega) - \hat{X}(\omega) \\ &= X(\omega) - G(\omega)[X(\omega) + V(\omega)] \\ &= [1 - G(\omega)]X(\omega) - G(\omega)V(\omega) \end{aligned} \quad (3.77)$$

We see that the error signal $e(t)$ is the superposition of the system $[1 - G(\omega)]$ acting on the signal $x(t)$, and the system $G(\omega)$ acting on the signal $v(t)$. Therefore, from Equation (3.75), we obtain

$$S_e(\omega) = [1 - G(\omega)][1 - G(-\omega)]S_x(\omega) - G(\omega)G(-\omega)S_v(\omega) \quad (3.78)$$

The variance of the estimation error is obtained from Equation (2.92) as

$$E[e^2(t)] = \frac{1}{2\pi} \int S_e(\omega) d\omega \quad (3.79)$$

To find the optimal filter $G(\omega)$ we need to minimize $E[e^2(t)]$, which means that we need to know $S_x(\omega)$ and $S_v(\omega)$, the statistical properties of the signal $x(t)$ and the noise $v(t)$.

3.4.1 Parametric filter optimization

In order to simplify the problem of the determination of the optimal filter $G(\omega)$, we can assume that the optimal filter is a first-order, low-pass filter (stable and causal³) with a bandwidth $1/T$ to be determined by parametric optimization.

$$G(\omega) = \frac{1}{1 + Tj\omega} \quad (3.80)$$

This may not be a valid assumption, but it reduces the problem to a parametric optimization problem. In order to simplify the problem further, suppose that $S_x(\omega)$ and $S_v(\omega)$ are in the following forms.

$$\begin{aligned} S_x(\omega) &= \frac{2\sigma^2\beta}{\omega^2 + \beta^2} \\ S_v(\omega) &= A \end{aligned} \quad (3.81)$$

In other words, the noise $v(t)$ is white. From Equation (3.78) we obtain

$$\begin{aligned} S_e(\omega) &= \left(\frac{Tj\omega}{1 + Tj\omega} \right) \left(\frac{-Tj\omega}{1 - Tj\omega} \right) \left(\frac{2\sigma^2\beta}{\omega^2 + \beta^2} \right) - \\ &\quad \left(\frac{1}{1 + Tj\omega} \right) \left(\frac{1}{1 - Tj\omega} \right) A \end{aligned} \quad (3.82)$$

Now we can substitute $S_e(\omega)$ in Equation (3.79) and differentiate with respect to T to find

$$T_{\text{opt}} = \frac{\sqrt{A}}{\sigma\sqrt{2\beta} - \beta\sqrt{A}} \quad (3.83)$$

■ EXAMPLE 3.8

If $A = \sigma = \beta = 1$ then the optimal time constant of the filter is computed as

$$\begin{aligned} T &= \frac{1}{\sqrt{2} - 1} \\ &\approx 2.4 \end{aligned} \quad (3.84)$$

and the optimal filter is given as

$$G(\omega) = \frac{1}{1 + j\omega T}$$

³A causal system is one whose output depends only on present and future inputs. Real-world systems are always causal, but a filter that is used for postprocessing may be noncausal.

$$\begin{aligned}
&= \frac{1/T}{1/T + j\omega} \\
g(t) &= \frac{1}{T} e^{-t/T} \quad t \geq 0
\end{aligned} \tag{3.85}$$

Converting this filter to the time domain results in

$$\dot{\hat{x}} = \frac{1}{T}(-\hat{x} + y) \tag{3.86}$$

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3.4.2 General filter optimization

Now we take a more general approach to find the optimal filter. The expected value of the estimation error can be computed as

$$\begin{aligned}
e(t) &= x(t) - \hat{x}(t) \\
e^2(t) &= x^2(t) - 2x(t)\hat{x}(t) + \hat{x}^2(t) \\
&= x^2(t) - 2x(t) \int g(u)[x(t-u) + v(t-u)] du + \\
&\quad \int \int g(u)g(\gamma)[x(t-u) + v(t-u)] \times \\
&\quad [x(t-\gamma) + v(t-\gamma)] du d\gamma \\
E[e^2(t)] &= E[x^2(t)] - 2 \int g(u)R_x(u) du + \\
&\quad \int \int g(u)g(\gamma)[R_x(u-\gamma) + R_v(u-\gamma)] du d\gamma
\end{aligned} \tag{3.87}$$

Now we can use a calculus of variations approach [Fom00, Wei74] to find the filter $g(t)$ that minimizes $E[e^2(t)]$. Replace $g(t)$ in the above equation with $g(t) + \epsilon\eta(t)$, where ϵ is some small number, and $\eta(t)$ is an arbitrary perturbation in $g(t)$. The calculus of variations says that we can minimize $E(e^2(t))$ by setting

$$\left. \frac{\partial E(e^2(t))}{\partial \epsilon} \right|_{\epsilon=0} = 0 \tag{3.88}$$

and thus solve for the optimal $g(t)$. From Equation (3.87) we can write

$$\begin{aligned}
R_e(0) &= R_x(0) - 2 \int [g(u) + \epsilon\eta(u)]R_x(u) du + \\
&\quad \int \int [g(u) + \epsilon\eta(u)][g(\gamma) + \epsilon\eta(\gamma)][R_x(u-\gamma) + R_v(u-\gamma)] du d\gamma
\end{aligned} \tag{3.89}$$

Taking the partial derivative with respect to ϵ gives

$$\begin{aligned}
\frac{\partial R_e(0)}{\partial \epsilon} &= -2 \int \eta(u) R_x(u) du + \\
&\quad \int \int [\eta(u)g(\gamma) + \eta(\gamma)g(u) + 2\epsilon\eta(u)\eta(\gamma)] \times \\
&\quad [R_x(u-v) + R_v(u-\gamma)] du d\gamma \\
\left. \frac{\partial R_e(0)}{\partial \epsilon} \right|_{\epsilon=0} &= -2 \int \eta(\tau) R_x(\tau) d\tau + \\
&\quad \int \int \eta(\tau)g(\gamma) [R_x(\tau-\gamma) + R_v(\tau-\gamma)] d\tau d\gamma + \\
&\quad \int \int \eta(\tau)g(u) [R_x(u-\tau) + R_w(u-\tau)] d\tau du \quad (3.90)
\end{aligned}$$

Now recall from Equation (2.87) that $R_x(\tau-u) = R_x(u-\tau)$ [i.e., $R_x(\tau)$ is even] if $x(t)$ is stationary. In this case, the above equation can be written as

$$\begin{aligned}
0 &= -2 \int \eta(\tau) R_x(\tau) d\tau + \\
&\quad 2 \int \int \eta(\tau)g(u) [R_x(u-\tau) + R_v(u-\tau)] d\tau du \quad (3.91)
\end{aligned}$$

This gives the necessary condition for the optimality of the filter $g(t)$ as follows:

$$\int \eta(\tau) \left[-R_x(\tau) + \int g(u) [R_x(u-\tau) + R_v(u-\tau)] du \right] d\tau = 0 \quad (3.92)$$

We need to solve this for $g(t)$ to find the optimal filter.

3.4.3 Noncausal filter optimization

If we do not have any restrictions on causality of our filter, then $g(t)$ can be nonzero for $t < 0$, which means that our perturbation $\eta(t)$ can also be nonzero for $t < 0$. This means that the quantity inside the square brackets in Equation (3.92) must be zero. This results in

$$\begin{aligned}
R_x(\tau) &= \int g(u) [R_x(u-\tau) + R_v(u-\tau)] du \\
&= g(\tau) * [R_x(\tau) + R_v(\tau)] \\
S_x(\omega) &= G(\omega) [S_x(\omega) + S_v(\omega)] \\
G(\omega) &= \frac{S_x(\omega)}{S_x(\omega) + S_v(\omega)} \quad (3.93)
\end{aligned}$$

The transfer function of the optimal filter is the ratio of the power spectrum of the signal $x(t)$ to the sum of the power spectrums of $x(t)$ and the noise $v(t)$.

■ EXAMPLE 3.9

Consider the system discussed in Example 3.8 with $A = \beta = \sigma = 1$. The signal and noise power spectra are given as

$$\begin{aligned} S_x(\omega) &= \frac{2}{\omega^2 + 1} \\ S_v(\omega) &= 1 \end{aligned} \quad (3.94)$$

From this we obtain the optimal noncausal filter from Equation (3.93) as

$$\begin{aligned} G(\omega) &= \frac{2}{\omega^2 + 3} \\ &= \frac{1}{\sqrt{3}} \left(\frac{2\sqrt{3}}{\omega^2 + 3} \right) \\ g(t) &= \frac{1}{\sqrt{3}} e^{-\sqrt{3}|t|} \\ &\approx 0.58 e^{-0.58|t|}, \quad t \in [-\infty, \infty] \end{aligned} \quad (3.95)$$

In order to find a time domain representation of the filter, we perform a partial fraction expansion of $G(\omega)$ to find the causal part and the anticausal⁴ part of the filter⁵:

$$G(\omega) = \underbrace{\frac{1}{\sqrt{3}(j\omega + \sqrt{3})}}_{\text{causal filter}} + \underbrace{\frac{1}{\sqrt{3}(-j\omega + \sqrt{3})}}_{\text{anticausal filter}} \quad (3.96)$$

From this we see that

$$\begin{aligned} \hat{X}(\omega) &= \frac{1}{\sqrt{3}(j\omega + \sqrt{3})} Y(s) - \frac{1}{\sqrt{3}(j\omega - \sqrt{3})} Y(s) \\ &= \hat{X}_c(\omega) + \hat{X}_a(\omega) \end{aligned} \quad (3.97)$$

$\hat{X}_c(\omega)$ and $\hat{X}_a(\omega)$ (defined by the above equation) are the causal and anticausal part of $\hat{X}(\omega)$, respectively. In the time domain, this can be written as

$$\begin{aligned} \hat{x}(t) &= \hat{x}_c(t) + \hat{x}_a(t) \\ \dot{\hat{x}}_c &= -\sqrt{3}\hat{x}_c + y/\sqrt{3} \\ \dot{\hat{x}}_a &= \sqrt{3}\hat{x}_a - y/\sqrt{3} \end{aligned} \quad (3.98)$$

The $\dot{\hat{x}}_c$ equation runs forward in time and is therefore causal and stable. The $\dot{\hat{x}}_a$ equation runs backward in time and is therefore anticausal and stable. (If it ran forward in time, it would be unstable.)

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⁴An anticausal system is one whose output depends only on present and future inputs.

⁵The MATLAB function RESIDUE performs partial fraction expansions.

3.4.4 Causal filter optimization

If we require a causal filter for signal estimation, then $g(t) = 0$ for $t < 0$, and the perturbation $\eta(t)$ must be equal to 0 for $t < 0$. In this case, Equation (3.92) gives

$$R_x(\tau) - \int g(u)[R_x(u - \tau) + R_v(u - \tau)] du = 0, \quad t \geq 0 \quad (3.99)$$

The initial application of this equation was in the field of astrophysics in 1894 [Sob63]. Explicit solutions were thought to be impossible, but Norbert Wiener and Eberhard Hopf became instantly famous when they solved this equation in 1931. Their solution was so impressive that the equation became known as the Wiener-Hopf equation.

To solve Equation (3.99), postulate some function $a(t)$ that is arbitrary for $t < 0$, but is equal to 0 for $t \geq 0$. Then we obtain

$$\begin{aligned} R_x(\tau) - \int g(u)[R_x(u - \tau) + R_v(u - \tau)] du &= a(\tau) \\ S_x(\omega) - G(\omega)[S_x(\omega) + S_v(\omega)] &= A(\omega) \end{aligned} \quad (3.100)$$

For ease of notation, make the following definition:

$$S_{xv}(\omega) = S_x(\omega) + S_v(\omega) \quad (3.101)$$

Then Equation (3.100) becomes

$$S_x(\omega) - G(\omega)S_{xv}^+(\omega)S_{xv}^-(\omega) = A(\omega) \quad (3.102)$$

where $S_{xv}^+(\omega)$ is the part of $S_{xv}(\omega)$ that has all its poles and zeros in the LHP (and hence corresponds to a causal time function), and $S_{xv}^-(\omega)$ is the part of $S_{xv}(\omega)$ that has all its poles and zeros in the RHP (and hence corresponds to an anticausal time function). Equation (3.102) can be written as

$$G(\omega)S_{xv}^+(\omega) = \frac{S_x(\omega)}{S_{xv}^-(\omega)} - \frac{A(\omega)}{S_{xv}^-(\omega)} \quad (3.103)$$

The term on the left side corresponds to a causal time function [assuming that $g(t)$ is stable]. The last term on the right side corresponds to an anticausal time function. Therefore,

$$\begin{aligned} G(\omega)S_{xv}^+(\omega) &= \text{causal part of } \frac{S_x(\omega)}{S_{xv}^-(\omega)} \\ G(\omega) &= \frac{1}{S_{xv}^+(\omega)} \left[\text{causal part of } \frac{S_x(\omega)}{S_{xv}^-(\omega)} \right] \end{aligned} \quad (3.104)$$

This gives the TF of the optimal causal filter.

■ EXAMPLE 3.10

Consider the system discussed in Section 3.4.1 with $A = \beta = \sigma = 1$. This was also discussed in Example 3.9. For this example we have

$$S_x(\omega) = \frac{2}{\omega^2 + 1}$$

$$\begin{aligned}
S_v(\omega) &= 1 \\
S_{xv}(\omega) &= \frac{\omega^2 + 3}{\omega^2 + 1}
\end{aligned} \tag{3.105}$$

Splitting this up into its causal and anticausal factors gives

$$\begin{aligned}
S_{xv}(\omega) &= \underbrace{\left(\frac{j\omega + \sqrt{3}}{j\omega + 1} \right)}_{S_{xv}^+(\omega)} \underbrace{\left(\frac{-j\omega + \sqrt{3}}{-j\omega + 1} \right)}_{S_{xv}^-(\omega)} \\
\frac{S_x(\omega)}{S_{xv}^-(\omega)} &= \frac{2(-j\omega + 1)}{(\omega^2 + 1)(-j\omega + \sqrt{3})} \\
&= \frac{2}{(-j\omega + \sqrt{3})(j\omega + 1)} \\
&= \underbrace{\frac{\sqrt{3} - 1}{j\omega + 1}}_{\text{causal part}} + \underbrace{\frac{\sqrt{3} - 1}{-j\omega + \sqrt{3}}}_{\text{anticausal part}}
\end{aligned} \tag{3.106}$$

Equation (3.104) gives

$$\begin{aligned}
G(\omega) &= \left(\frac{j\omega + 1}{j\omega + \sqrt{3}} \right) \left(\frac{\sqrt{3} - 1}{j\omega + 1} \right) \\
&= \frac{\sqrt{3} - 1}{j\omega + \sqrt{3}} \\
g(t) &= (\sqrt{3} - 1)e^{-\sqrt{3}t}, \quad t \geq 0
\end{aligned} \tag{3.107}$$

This gives the TF and impulse response of the optimal filter when causality is required.

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3.4.5 Comparison

Comparing the three examples of optimal filter design presented in this section (Examples 3.8, 3.9, and 3.10), it can be shown that the mean square errors of the filter are as follows [Bro96]:

- Parameter optimization method: $E[e^2(t)] = 0.914$
- Causal Wiener filter: $E[e^2(t)] = 0.732$
- Noncausal Wiener filter: $E[e^2(t)] = 0.577$

As expected, the estimation error decreases when we have fewer constraints on the filter. However, the removal of constraints makes the filter design problem more difficult. The Wiener filter is not very amenable to state estimation because of difficulty in extension to MIMO problems with state variable descriptions, and difficulty in application to signals with time-varying statistical properties.